Ming Gong, Ph.D.

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SUMMARY

Hands-on **applied-AI engineering leader** with **13+ years** across the full frontier-model production stack — from pre-training-era cross-lingual encoders (Unicoder, EMNLP 2019) and **CodeBERT** (3,000+ citations) through BERT/DNN compression and distillation for production serving, to LLM post-training, SLM fine-tuning, generative answer systems, and agentic search shipped to **global customers**. Currently **Head of Machine Learning Engineering, Knowledge AI** at Atlassian — leading search, retrieval, knowledge graph, generative answers, and agentic search across the Atlassian Suite, accountable for an extended org of **130+**.

Previously **Group Senior Principal Applied Sciences Manager at Microsoft** (11 years), taking Bing Question-Answering and Copilot Search from zero to global and shipping **Copilot Search one month before Google's AI Mode**. Operates end-to-end across frontier-model research, eval and quality methodology, production inference and serving, and 0→1 delivery — uniquely multi-dimensional in **applied sciences & full-stack engineering**, **AI products & research** (50+ publications, 11,000+ citations, 20+ patents), and **consumer & enterprise products**. Two-time Microsoft MLADS Distinguished Contribution Award recipient (first-ever back-to-back); Best Paper Award, IJCAI '24 AI4Research.

TECHNICAL HIGHLIGHTS

‣ Frontier Model Work

- **Distillation & compression for production serving**: Authored novel model compression that powered the first BERT model deployed in Bing — within 1 month of BERT open-source, zero added GPU cost; +3.0% answer coverage, +5.0% precision. Later built **Glow**, a distillation framework producing task-specific SLMs that replaced GPT-4-class models in production — **first-token latency 20-30s → ~5ms (~4000× reduction)**, scaled to 5-10% of Bing global query coverage.
- **Universal multilingual model architecture**: Designed a cross-lingual model serving 100+ languages on ~13% of the GPU footprint a per-language approach would have required. Co-authored **Unicoder** (EMNLP/IJCNLP 2019), one of the earliest universal multilingual encoders, and **CodeBERT** (EMNLP Findings 2020, 3,000+ citations), the first pre-trained model for programming + natural languages.
- **LLM post-training & SLM fine-tuning at production scale**: Established end-to-end post-training and fine-tuning framework for Copilot Search (multi-turn AI search vertical), including R1-distilled task-specific models; <5s first-token latency in production at consumer scale.
- **Retrieval architecture innovation**: Replaced industry-standard sliding-window passage indexing with runtime **query-dependent passage extraction** via GNN document modeling + multi-teacher distillation to 40ms latency. Held **#1 on Google Natural Questions benchmark for ~1 year**; architecture adopted across Bing QnA, Knowledge, People Also Ask, and Azure Custom QnA.

‣ Eval & Quality Methodology

- **Tier-wise hallucination control framework** for LLM-generated search results — balanced capability and grounding requirements across content tiers. Held the technical position through 5-6 leadership reviews against split organizational consensus; framework subsequently adopted by Edge Copilot and Office Copilot.
- **Systematic blinded human side-by-side (SBS) evaluation**: 10+ DNN ranker releases replacing legacy GBDT (400+ hand-crafted features), each win-validated against Google via blinded human preference; cumulative wins drove Bing QnA from competitive-loss to competitive-parity on multiple verticals.
- **Diagnosis-first hands-on intervention**: For entity linking quality regression (40% → 90%+ precision), invested 2 months in hands-on data and model diagnosis before redirecting technical approach — illustrates eval-driven iteration over surface-level fixes.

‣ Production Inference & Serving

- **NRT (near-real-time) inference pipeline** for Bing QnA: co-owned design with 8 platform teams across time zones; pipeline became the platform team's flagship demo and blueprint for subsequent products.
- **Generative Answers on Bing SERP**: 20% query coverage globally, +2.0% SBS lift via streaming generative answers with click-through to Bing Chat.
- **Copilot Search**: multi-turn AI search vertical unifying GenQnA, GenSERP, and Magazine; shipped **April 2025, one month before Google's AI Mode**.

‣ Multilingual & Cross-Lingual Scaling

- Scaled Bing QnA from English-only to **100+ languages and 230+ markets** via systematic cross-lingual methodology. Placed the universal-multilingual bet ~1 year before globalization became official org strategy.
- **2× Microsoft MLADS Distinguished Contribution Award** (CTO Office global award; first-ever back-to-back recipient in conference history; <2 / 300+ acceptance rate).
- Authored two large-scale survey tutorials: *Language Scaling: Applications, Challenges and Approaches* (KDD '21) and *Scaling out NLP Applications to 100+ Languages* (WWW '21).

‣ Research Impact

- **Best Paper Award, IJCAI '24 AI4Research** — *Step-Back Profiling: Distilling User History for Personalized Scientific Writing*.
- **CodeBERT** (EMNLP Findings 2020): 3,000+ citations, foundational pre-trained model for code intelligence.
- 50+ publications across NeurIPS-tier venues (ACL, EMNLP, NAACL, KDD, AAAI, SIGIR, ICLR, WSDM, CIKM, COLING), 11,000+ citations on Google Scholar, 20+ patents granted.

EXPERIENCE

Atlassian **Feb 2025 – Present**
Head of Machine Learning Engineering, Knowledge AI *Bellevue, WA*

Lead **Knowledge AI** at Atlassian — search, retrieval, knowledge graph, generative answers, and agentic search across Jira, Confluence, and Rovo. Hands-on technical leadership across ranking model architecture, RAG pipelines, agent framework design, generative UI, and LLM post-training methodology. Grew the IC org from 30 to 60+ (70% scientists, 30% engineers); recruited 2 Principal Managers and 3 Principal Tech Leads; established greenfield ML engineering sites in Canada and Singapore.

- **Jira Search rebuild (MVP in 4 months, full launch in 12)**: Rebuilt the ranking stack and search UX from the ground up; drove **Search MAU 2.5M → 8.5M (3.4×)**, helping AI Org exceed Search MAU L1 OKR. Premiered at Team '25. Cross-org V-Team scaled from ~30 to 115 contributors across 8 partner orgs; established a Coalition Council for cross-org technical alignment.
- **Generative Answers in Search + AI Definitions**: novel generative UI framework drove **sustained AI MAU 100K → 1.2M**; technical work spans retrieval grounding, citation correctness, streaming generation, and latency optimization.
- **Knowledge fundamentals (collaboration graph, entity linking, entity recommendation)**: now adopted by multiple product teams for personalized features. Led entity linking turnaround from **40% to 90%+ precision** — 2 months hands-on diagnosis before redirecting approach; protected team stability through parallel senior hiring.
- **Jira Agents in Rovo Chat**: shipped within 3 months; code-driven agent architecture **beat Manus and GenSpark on the GAIA benchmark in early 2026**; primary driver of Rovo adoption.
- **Lumina — generative UI layer for enterprise GenAI Search**: identified a cross-product platform gap no team was chartered to own; POC in <1 month, MVP shipped early 2026. **+20% engagement uplift, +30% SBS lift vs. competitor AI experiences**; adopted as Atlassian's standard generative presentation framework, integrating across all GenAI products. Previewed at Team '25 EU AI Super Session; live demo at Team '26 Founder Keynote.

Microsoft — Search Technology Center Asia (Bing, Edge, Copilot) **Aug 2013 – Dec 2024**
Group Senior Principal Applied Sciences Manager (2019–2024) · Senior Dev Manager (2018–2019) · Senior Tech Lead (2016–2018) · SWE II (2013–2015) *Beijing, China*

Built and led a team of 40+ engineers and applied scientists from scratch across Beijing and Suzhou (70% scientists, 30% engineers; 5 managers), spanning Bing Search, Copilot, Cross-Lingual QA, Knowledge Graph, Recommendation, and Edge browser AI.

Frontier model & inference work

- **Bing Question-Answering, zero-to-global, traditional ML → DL → pre-trained models → LLM (2016–2024)**: built Bing QnA ground-up and led its evolution through every major AI paradigm shift. Authored the BERT compression method that powered Bing's first BERT deployment within 1 month of open-source, zero added GPU cost — +3.0% coverage, +5.0% precision. Universal multilingual model served 100+ languages on ~13% of a per-language approach's GPU footprint. **2× MLADS Distinguished Contribution Award (back-to-back, first ever)**.
- **Generative SERP — distillation & hallucination control (2023–2024)**: built **Glow**, a distillation framework producing task-specific SLMs that replaced GPT-4-class models in production — first-token latency 20-30s → ~5ms, scaled to 5-10% of Bing global query coverage. Proposed tier-wise hallucination control framework against split org consensus; held position through 5-6 leadership reviews; framework adopted by Edge Copilot and Office Copilot.
- **Copilot Search (Bing AI vertical, 2024–2025)**: multi-turn AI search experience unifying GenQnA, GenSERP, and Magazine; built scalable LLM generation pipelines and task-specific SLMs (incl. R1-distilled). **Shipped April 2025 — one month before Google's AI Mode**.
- **Query-Dependent Passage Extraction (2019–2022)**: replaced industry-standard sliding-window passage indexing with runtime query-dependent extraction via GNN document modeling + multi-teacher distillation to 40ms latency. **Held #1 on Google Natural Questions benchmark for ~1 year**; architecture adopted across Bing QnA, Knowledge, People Also Ask, and Azure Custom QnA.

Production product work at consumer scale

- **Knowledge Card for 150M Entities (2021–2023)**: multi-modal entity aggregation via template + MRC DL extraction and LLM generation. **60% Bing query traffic, 65M DAU, +3.0% SBS lift**; certified as a Bing "WoW" feature (<10 out of 100+ features).
- **Generative Answers on Bing SERP (2023–2024)**: 20% query coverage on Bing global markets, +2.0% SBS lift via streaming generative answers and click-through to Bing Chat.
- **NLU Platform "Orca" (2020–2025)**: unified query understanding pipeline (domain, intent, slots, topics) powering answer triggering. 95%+ of Bing answers (coverage ≥ 0.05%) onboarded; 60% of onboarded answers closed gap with Google on SBS.

- **Microsoft Copilot on Edge (2023–2024)**: long-document summarization and chat over PDFs and long-form web pages — **1000+ page docs at 95%+ accuracy**; demoed at Microsoft Copilot 2023 event; shipped to Edge and Windows Copilot.
- **Underside (Edge right-rail Insight Pane, 2022–2024)**: AI-powered right-rail panel featuring page summary, topics, QnA, related search, related pages. **Edge engaged DAU +10%, Search DAU +1.2M**.
- **Segment entity search & recommendation (2021–2024)**: shipped recipe, job, and movie answers — beat Google on relevance and SBS on recipe and job; closed SBS gap with Google on movie recommendation.

People & community

- Founded and led **Microsoft AI School APRD (2017–2024)** — 8-year regional AI education program; 10K+ participants, 2K+ certified graduates; the most influential AI training program in Microsoft Asia-Pacific R&D.

INDEPENDENT TECHNICAL WORK

- **X-SuperAgent** (2025–): an Atlassian-compliant implementation of the OpenClaw multi-agent framework; integrates with first-party (1P) and third-party (3P) workplace apps to facilitate productivity improvements across day-to-day work scenarios.
- **VoicePilot** (2025): a cross-platform voice-input application (iOS, Windows, macOS) that markedly improves the efficiency of interacting with AI tools — rapidly distilling lengthy verbal input into clear, well-structured prompts.

EDUCATION

Ph.D., Graphics and Visual Computing — Institute of Computing Technology, Chinese Academy of Sciences **2008 – 2013**

SELECTED AWARDS

- **Best Paper Award, IJCAI '24 AI4Research** — *Step-Back Profiling: Distilling User History for Personalized Scientific Writing*.
- **2021 Fall Microsoft MLADS Distinguished Contribution Award** (2 / 250+; CTO Office global award).
- **2021 Spring Microsoft MLADS Distinguished Contribution Award** (2 / 300+) — broke conference history as the first ever back-to-back recipient.
- **2023 Microsoft Global Hackathon — Edge Theatre**: Best Overall Hack; Greater China Region: Most Popular Hack.
- **2023 Microsoft Global Hackathon — WebXT China, DeKG**: Best Innovativeness & Originality Hack.
- Microsoft AI Core Q3 FY18 Greatness Award.

SELECTED PUBLICATIONS (50+ peer-reviewed papers, 11,000+ citations on Google Scholar — full list on scholar profile)

‣ Code Intelligence

- Zhangyin Feng, Daya Guo, Duyu Tang, Nan Duan, Xiaocheng Feng, **Ming Gong**, Linjun Shou, Bing Qin, Ting Liu, Daxin Jiang, Ming Zhou. "CodeBERT: A Pre-Trained Model for Programming and Natural Languages." *EMNLP Findings* 2020. (3,000+ citations)
- Junjie Huang, Duyu Tang, Linjun Shou, **Ming Gong**, et al. "CoSQA: 20,000+ Web Queries for Code Search and Question Answering." *ACL* 2021.

‣ LLM, Agentic AI & Multimodal Pre-training

- Yaobo Liang, Chenfei Wu, Ting Song, et al., **Ming Gong**, Nan Duan. "TaskMatrix.AI: Completing Tasks by Connecting Foundation Models with Millions of APIs." *Intelligent Computing* 2023.
- Gen Li, Nan Duan, Yuejian Fang, **Ming Gong**, Daxin Jiang. "Unicoder-VL: A Universal Encoder for Vision and Language by Cross-modal Pre-training." *AAAI* 2019.
- Nuo Chen, Ning Wu, Shining Liang, **Ming Gong**, et al. "Is Bigger and Deeper Always Better? Probing LLaMA Across Scales and Layers." *arXiv* 2023.
- Ning Wu, **Ming Gong**, Linjun Shou, et al. "Large Language Models are Diverse Role-Players for Summarization Evaluation." *NLPCC* 2023.

‣ Model Training, Compression & Distillation

- **Ming Gong**, Linjun Shou, Wutao Lin, et al. "NeuronBlocks — Building Your NLP DNN Models Like Playing Lego." *EMNLP/IJCNLP* 2019.
- Ze Yang, Linjun Shou, **Ming Gong**, Wutao Lin, Daxin Jiang. "Model Compression with Two-stage Multi-teacher Knowledge Distillation for Web Question Answering System." *WSDM* 2020.
- Fei Yuan, Linjun Shou, Jian Pei, Wutao Lin, **Ming Gong**, Daxin Jiang. "Reinforced Multi-Teacher Selection for Knowledge Distillation." *AAAI* 2021.

‣ Cross-Lingual & Multilingual Modeling

- H. Huang, Yaobo Liang, N. Duan, **Ming Gong**, et al. "Unicoder: A Universal Language Encoder by Pre-training with Multiple Cross-lingual Tasks." *EMNLP/IJCNLP* 2019.
- Nuo Chen, Linjun Shou, **Ming Gong**, Jian Pei, Daxin Jiang. "Bridging the Gap between Language Models and Cross-Lingual Sequence Labeling." *NAACL* 2022.
- Nuo Chen, Linjun Shou, **Ming Gong**, Jian Pei. "From Good to Best: Two-Stage Training for Cross-lingual Machine Reading Comprehension." *AAAI* 2022.
- Shining Liang, **Ming Gong**, Jian Pei, Linjun Shou, Daxin Jiang. "Reinforced Iterative Knowledge Distillation for Cross-Lingual Named Entity Recognition." *KDD* 2021.

- Nuo Chen, Linjun Shou, Tengtao Song, **Ming Gong**, et al. "Structural Contrastive Pretraining for Cross-Lingual Comprehension." *ACL* 2023.
- Nuo Chen, Zinan Zheng, Ning Wu, **Ming Gong**, et al. "Breaking Language Barriers in Multilingual Mathematical Reasoning." *arXiv* 2023.
- Linjun Shou, **Ming Gong**, Jian Pei, et al. "Language Scaling: Applications, Challenges and Approaches." *KDD Tutorial* 2021.
- Linjun Shou, **Ming Gong**, Jian Pei, et al. "Scaling out NLP Applications to 100+ Languages." *WWW Tutorial* 2021.

‣ Dense Retrieval & RAG

- Shengyao Zhuang, Linjun Shou, Jian Pei, **Ming Gong**, et al. "Typos-aware Bottlenecked Pre-Training for Robust Dense Retrieval." *SIGIR* 2023.
- Houxing Ren, Linjun Shou, Jian Pei, Ning Wu, **Ming Gong**, Daxin Jiang. "Lexicon-Enhanced Self-Supervised Training for Multilingual Dense Retrieval." *EMNLP* 2023.
- Shunyu Zhang, Yaobo Liang, **Ming Gong**, Daxin Jiang, Nan Duan. "Modeling Sequential Sentence Relation to Improve Cross-lingual Dense Retrieval." *ICLR* 2023.
- Shunyu Zhang, Yaobo Liang, **Ming Gong**, Daxin Jiang, Nan Duan. "Multi-View Document Representation Learning for Open-Domain Dense Retrieval." *ACL* 2022.
- Linjun Shou, S. Bo, Fei-Xiang Cheng, **Ming Gong**, et al. "Mining Implicit Relevance Feedback from User Behavior for Web Question Answering." *KDD* 2020.

‣ Knowledge & Entity Understanding

- Zilin Xiao, **Ming Gong**, Jie Wu, et al. "Instructed Language Models with Retrievers Are Powerful Entity Linkers." *EMNLP* 2023.
- Zilin Xiao, Linjun Shou, Xingyao Zhang, Jie Wu, **Ming Gong**, et al. "Coherent Entity Disambiguation via Modeling Topic and Categorical Dependency." *EMNLP Findings* 2023.

‣ Personalization & Scientific Writing

- *(Best Paper, IJCAI '24 AI4Research)* "Step-Back Profiling: Distilling User History for Personalized Scientific Writing."

SELECTED PATENTS (20+ granted; selection below)

- Model Generation based Model Compression. 2019. (MS#406805-CN-NP)
- Model Selection Learning for Knowledge Distillation. 2020. (MS#408426-CN-NP)
- Multi-Model Joint Denoising Training. 2021. (MS#409566-CN-NP)
- Lexicon-Enhanced Self-Supervised Training for Multilingual Dense Retrieval. 2022. (MS#412496-CN-NP)
- Sentence Representation Generation for Cross-lingual Retrieval. 2022.
- Representation Learning of Cross-Language Texts. 2021. (MS#409565-CN-NP)
- Cross-Lingual Task Training. 2019. (MS#406497-CN-NP)
- From Good to Best: Two-Stage Training for Cross-lingual Machine Reading Comprehension. 2021.
- Contrastive Learning for Question Answering. 2020. (MS#407988-CN-NP)
- Providing QA Training Data and Training a QA Model based on Implicit Relevance Feedbacks. 2020. (MS#407927-CN-NP)
- Question Generation from Queries. 2021. (MS#410241-CN-NP)
- Assertion-based Question Answering with Question-Aware Open Information Extraction. 2019.
- Representation Learning of Semi-structured Data. 2020. (MS#408908-CN-NP)
- Graph based Fact Clustering for Diversity and Freshness. 2021.
- Controversial Poll Auto Generation by Generation Model and Crowdsourcing. 2021.
- Personalized Multilingual Content Recommendation based on Robust Feature Network. 2023. (MS#412659-CN-NP)
- Sequential Recommendation based on Cross-Domain Behavior Data. 2023.
- Sequential Recommendation Based on Dual Contrastive Interest Learning. 2022.
- Temporal Co-Contrastive Learning-Based Node Representation Generation. 2022.
- Content Recommendation Based on Graph Enhanced Collaborative Filtering. 2022.
- Knowledge-enhanced Content-Aware Collaborative Filtering for Recommendation. 2021.
- Hierarchical Representation Learning of User Interest. 2021. (MS#410462-CN-NP)